

EXPERIMENT - 14

Brute Force Attack Study in Kali Linux

Step 1: Start Environment

Open Oracle VirtualBox and start Kali Linux.

Step 2: Setup Test Environment

Use a local test account with dummy credentials in an isolated environment.

Step 3: Study Attack Concept

Understand how repeated login attempts are used to guess passwords.

Step 4: Simulated Demonstration

Observe multiple authentication attempts and system responses in a controlled setup.

Step 5: Analyze Risks and Prevention

Identify risks of weak passwords and study prevention methods like strong passwords, account lockout, and multi-factor authentication.

Output

Simulated login attempts demonstrate how weak passwords can be guessed.

```
nu@saira-banu-VirtualBox:~/zphisher$ hydra -1 testuser -p pass.txt
.1
.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use
r secret service organizations, or for illegal purposes (this is n
ese *** ignore laws and ethics anyway).

nvalid option -- '1'
nu@saira-banu-VirtualBox:~/zphisher$ hydra -l testuser -p pass.txt
.1
.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use
r secret service organizations, or for illegal purposes (this is n
ese *** ignore laws and ethics anyway).

https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-26

] Many SSH configurations limit the number of parallel tasks, it f
to reduce the tasks: use -t 4
ax 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1

ttacking ssh://127.0.0.1:22/
arget completed, 0 valid password found
https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-26

nu@saira-banu-VirtualBox:~/zphisher$
```

Result

Brute force attack concepts were successfully studied, highlighting the importance of strong authentication mechanisms.